

Markup Languages

Markdown, HTML, and L^AT_EX

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Introduction

- ▶ **Markup Languages?** Computer languages used to format text
- ▶ **Format?** You'll see the following:
 - ▶ Definitions of language constructs
 - ▶ Basic syntax and behavior
 - ▶ Examples with outputs
- ▶ **Why Markup?** You'll learn how to structure the appearance of your writing. In particular, nice-looking assignment reports.
(Also, almost all Math and CSE research papers use $\text{\LaTeX}$.)
- ▶ **These Slides?** See our [GitHub](#) repository.
- ▶ **Coding Background?** No background needed for this topic.

Any questions?



Outline

1 Markdown

- Introduction to Markdown
- Why Use Markdown?
- Common Syntax

2 HTML

3 $\text{\LaTeX}$

- Document Structure
- Simple Markup
- Environments
- Useful Commands
- Math Mode



Introduction to Markdown

- ▶ Lightweight markup language with easy-to-read syntax.
- ▶ Converts plain text into HTML and other formats.
- ▶ Why Use Markdown?

Why Use Markdown?

- ▶ Readability: Simple for anyone to read or write.
- ▶ Versatile: Used in documentation, notes, and web content.
- ▶ Widely Supported: Works with GitHub, VSCode, Jupyter, and more.

Common Syntax

- ▶ Headers: #
 - ▶ # for Header 1, ## for Header 2, ...
 - ▶ ##### for Header 6!
- ▶ Lists:
 - ▶ * for bullet list
 - ▶ 1., 2., ... for numbered list
- ▶ `*text*` for bold **text**, `_text_` for italic *text*

- ▶ Hyper Text Markup Language
- ▶ Is standard for creating Web pages
- ▶ Consists of a series of elements
- ▶ Elements tell the browser how to display the content
- ▶ Elements label pieces of content such as “this is a heading”, “this is a paragraph”, “this is a link”, etc.

Demo time!

- ▶ Derivative of Donald Knuth's T_EX, made by Leslie Lamport.
- ▶ Designed for print documents—output traditionally was a .dvi file, but later, .ps and .pdf became possible.
- ▶ Huge amounts of community extensions for L^AT_EX in the form of packages. See the *CTAN* (Comprehensive T_EX Archive Network).

Document Structure

Two parts:

- 1 Preamble: Command/macro definitions, package imports, and configurations. Example:

```
\documentclass[11pt]{article}
\usepackage{graphicx, float}
\def\myname{Mohammed}
```

- 2 Document: The actual formatted text. Example:

```
\begin{document}
  Hello! I am \myname.
\end{document}
```

Simple Markup

Some constructs for simple mark up:

- ▶ `\textit` for *italics*. $\leftarrow$ “`\textit{italics}`”
- ▶ `\textbf` for **boldface**. $\leftarrow$ “`\textbf{boldface}`”
- ▶ `\texttt` for typewriter. $\leftarrow$ “`\texttt{typewriter}`”
- ▶ To ignore commands, use `\verb`. $\leftarrow$ “`\verb|\verb|`”

Exercise. How did I typeset “`\verb|\verb|`”?

Environments

Environments use `\begin{}` and `\end{}`. Inside the scope of an environment, $\text{\LaTeX}$ behaves differently, and new commands become available. Examples:

- ▶ `itemize`: unordered lists.
- ▶ `enumerate`: ordered lists.
- ▶ `multicol`: multiple columns. (from `multicols` package.)
- ▶ `tabular`: tables.
- ▶ `verbatim`: multi-line `\verb`.
- ▶ `center` ..?
- ▶ `figure` ..! (recommended: use `float` package with this)

Useful Commands

- ▶ `\section`, `\subsection`, `\subsubsection`.
Automatically numbered!
 - ▶ `\tableofcontents` uses these to typeset a full table of contents.
 - ▶ `\today` gives the current date: November 14, 2024.
Good for cover pages!
 - ▶ `\includegraphics` displays an image:  (`graphicx` package.)
 - ▶ `\vskip`

and `\vspace` for vertical spacing.
 - ▶ `\hrule` to draw a horizontal line across the line width:
-

Math Mode

Use `\(` and `\)` for inline math, and `\[` and `\]` for display math.

► `\(x^2=4, x>0\)` gives $x^2 = 4, x > 0$

► `\[ \int_{-4}^5 \left( \frac{x^3}{3} + 7 \right) dx \]` gives:

$$\int_{-4}^5 \left(\frac{x^3}{3} + 7 \right) dx$$

► There's a lot here—math notation has a lot of symbols, and their names aren't *always* obvious, but not much can be done about that.

That's All, Folks!

Thank you!

Questions? Comments? Concerns?

